DATA ITEM DESCRIPTION

Title: SYSTEM/DESIGN TRADE STUDY REPORT

Number: DI-SESS-82438
AMSC Number: N10462
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20240222
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2024-006

Use/Relationship: The Trade Study Report used to document the decision rationale for designated trade studies. Trade studies involve a variety of engineering techniques including complete or partial simulation of operations. Trade studies accomplished at different level and reporting data may vary considerably. Trade study reports will document the results of system effectiveness, life cycle cost and risk analysis.

This Data Item Description (DID) contains the format, content, and intended use for the data product resulting from the work task described in the contract Statement of Work/Performance Work Statement.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. Contractor Format.

3. Contents. The Trade Study Report shall be organized into the following sections:

3.1. Scope. This section shall be divided into the following sub-sections:

3.2. Identification and listing of Functional and Technical Design Requirements for Tradeoff.

3.2.1. Functional Requirements are listed first.

3.2.2. Related Technical Requirements.

3.3. Identification of possible Design Approaches and Then Design Characteristics.

3.3.1. List possible approaches and identify significant design characteristics for each possible design.

3.3.2. The list generated concurrently with above requirements.

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3.3.3. Attainable design approaches pursued considering technical capabilities, cost, time schedules, system effectiveness, risks, and resource limitations or other constraints as specified in system requirement documentation.

3.3.4. List each potential and alternative design approach and identify significant characteristics applicable to each.

3.3.4.1. The Characteristics shall relate and be restrictive to the attributes of the design approach, which bears most directly on its feasibility in relation to the requirements.

3.3.4.2. The significant characteristics shall reflect predicted impact on such factors as cost, schedules, performance, survivability, safety, growth potential, maintainability, reliability, personnel and training requirements, facilities, security, vulnerability, transportability, procurability, and producibility.

3.4. Comparison of Matrix Design Approaches.

3.4.1. The matrix shall compare the design characteristics for each design approach to determine the degree to which the design approaches satisfy the functional and technical requirements.

3.4.2. The objective is to facilitate rapid comparison and evaluation of potential design approaches and to allow preliminary screening out those design approaches that are inconsistent with the functional and technical requirements.

3.5. Selection of Design Approach.

3.5.1. Select the most promising design approach and provide reasons to substantial the selection.

3.5.2. Provide rationale for the selection in the form of schematic diagrams, outline drawings, interface details, functional diagrams, reliability data, statistic interference data, narrative and any other backup data to support the selection.

3.5.3. The rationale shall cover the requirements the selected approach imposes on other areas of the systems, facilities, training, training equipment, human performance, and procedural data.

3.6. Reliability Development Growth Test.

3.6.1. List the subsystems selected for analysis and a description of the process and criteria used to select those subsystems.

3.6.2. For each subsystem selected, describe the type(s) of test selected for analysis and a description of the process and criteria used to make those selections.

3.6.3. For each individual test analyzed, provide an expected cost/benefit summary in addition to a detailed description of the analysis used to generate that summary.

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3.7. Trade Study Report Index.

3.7.1. The Trade Study Report Index shall identify all trade studies required, studies completed, and those to be completed.

End of DI-SESS-82438

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